
Uni-mHC: Unistochastic Manifold-Constrained Hyper-Connections with Quantum-Interference Disentanglement

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Abstract

Manifold-constrained hyper-connections (mHC) stabilize deep multi-stream residual training by projecting the inter-stream mixing matrix onto the Birkhoff polytope of doubly stochastic matrices via Sinkhorn–Knopp iterations. Entry-wise non-negativity, however, restricts mixing to *additive convex combinations*: features can never be subtracted, a limitation recently criticized in the signed-hyper-connections (sHC) line of work. We propose **Uni-mHC**, which lifts the constraint manifold from the Birkhoff polytope to the *unistochastic* set: a complex Cayley transform of a skew-Hermitian parameter matrix yields a unitary U whose modulus square $S = |U| \odot |U|$ is doubly stochastic *by construction*—no projection, no iteration. Macroscopically S preserves exact energy conservation; microscopically a heterodyne readout $W = \text{Re}(Ue^{i\phi})$ admits negative effective weights through destructive (phase- π) interference while remaining row-wise non-expansive. On a controlled subtractive task ($Y = X_0 - X_1$), non-negative baselines plateau at the theoretical floor $\frac{1}{2} \text{Var}(Y) \approx 0.99$ whereas Uni-mHC converges to 1.1×10^{-11} ; the operator is also the fastest and constraint-exact of all compared parameterizations, and the identity-initialized four-stream topology—initiated here with Uni-mHC—reaches $2.3\times$ lower validation loss than an identically trained vanilla GPT.

1 Introduction

Residual streams are the backbone of modern deep networks, from the two-path residual connection (He et al., 2016) to hyper-connections (HC) (Lu et al., 2025), which maintain n parallel streams with a learned mixing matrix at every block boundary. Unconstrained mixing can amplify signal exponentially across depth, so mHC (DeepSeek, 2025) constrains it to the Birkhoff polytope of doubly stochastic matrices (Sinkhorn, 1964; Idel and Wolf, 2015).

The Birkhoff constraint carries an expressivity tax: since $H_{jk} \geq 0$, every stream update is a convex combination—mixing can only *add*, never *subtract*. The sHC critique (sHC, 2026) argues that this purely additive coupling pushes deep stacks toward identity degeneration; yet simply allowing negative entries abandons the energy-conservation guarantees that motivated mHC. We resolve this dichotomy by moving from real to complex manifolds (Fig. 1): a complex Cayley transform gives a unitary U whose $S = |U|^{\odot 2}$ is *exactly* doubly stochastic, while a heterodyne readout $W = \text{Re}(U \text{diag } e^{i\phi})$ restores subtractive arithmetic through phase interference. Contributions: (i) an exact unistochastic operator, with proofs of doubly stochasticity and unconditional well-conditioning; (ii) an impossibility floor $\frac{1}{2} \text{Var}(Y)$ for all entrywise non-negative mixers, and an exact subtraction theorem for ours; (iii) operator benchmarks, the disentanglement experiment, and a four-stream nanoGPT integration (Sec. 3).

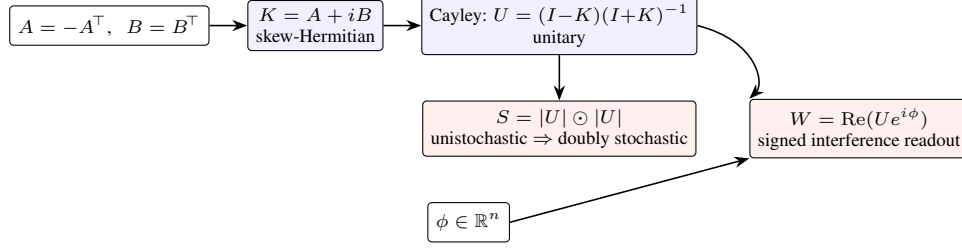


Figure 1: **Uni-mHC operator.** The complex Cayley transform of a skew-Hermitian K is unitary; $S = |U|^{\odot 2}$ is the doubly stochastic energy bookkeeping, while the heterodyne readout W is the signed mixing applied to streams.

2 Mathematical Theory

Setup. A hyper-connected network maintains n residual streams ($n = 4$ here); at every block boundary a mixing matrix $M \in \mathbb{R}^{n \times n}$ redistributes streams. mHC requires the mixer’s energy bookkeeping matrix to be doubly stochastic so that no stream’s energy is amplified or leaked.

Definition 1 (Uni-mHC operator). *Let $\theta_A, \theta_B \in \mathbb{R}^{n \times n}$ be free parameters and $\phi \in \mathbb{R}^n$ per-stream heterodyne phases. Set*

$$A = \theta_A - \theta_A^\top, \quad B = \theta_B + \theta_B^\top, \quad K = A + iB, \quad (1)$$

and define the unitary and the two mixing matrices

$$U = (I - K)(I + K)^{-1}, \quad S = \text{Re}(U)^{\odot 2} + \text{Im}(U)^{\odot 2}, \quad W = \text{Re}(U \text{diag}(e^{i\phi})). \quad (2)$$

S is the (unistochastic) energy matrix; W is the effective signed mixing applied to the streams.

Theorem 1 (Exact unistochasticity). *For any θ_A, θ_B : (i) K in (1) is skew-Hermitian; (ii) $I + K$ is invertible and U in (2) is unitary; (iii) $S = |U|^{\odot 2}$ is doubly stochastic; (iv) every singular value of $I + K$ is at least 1, so the Cayley solve is unconditionally well-conditioned.*

Proof. (i) $K^H = \overline{A + iB}^\top = A^\top - iB^\top = -A - iB = -K$. (ii) K is normal with purely imaginary spectrum $i\lambda_1, \dots, i\lambda_n$, $\lambda_j \in \mathbb{R}$; hence $I + K$ has eigenvalues $1 + i\lambda_j \neq 0$. Since $(I + K)(I - K) = I - K^2 = (I - K)(I + K)$, the inverses commute, and $UU^H = (I - K)[(I + K)^H]^{-1}(I - K)^H = (I - K)(I - K)^{-1}(I + K)^{-1}(I + K) = I$, using $(I + K)^H = I - K$; likewise $U^H U = I$. (iii) $\sum_k |U_{jk}|^2 = (UU^H)_{jj} = 1$ and $\sum_j |U_{jk}|^2 = (U^H U)_{kk} = 1$, and $S \geq 0$ entrywise: S is unistochastic, hence doubly stochastic. (iv) Singular values of $I + K$ are $|1 + i\lambda_j| = \sqrt{1 + \lambda_j^2} \geq 1$. $\square$

Unlike SK projection, which needs T iterations and leaves a residual column-sum error (measured 5×10^{-3} after $T = 20$, Sec. 3.1), Theorem 1 holds exactly by construction, and the operator costs one $n \times n$ complex linear solve; the Cayley map is a diffeomorphism onto a dense open subset of $U(n)$, so the parameterization is lossless and smooth, and gradients (through `torch.linalg.solve` in `complex64`) were finite in all experiments.

Proposition 1 (Non-expansive heterodyne mixing). *Writing $U_{jk} = |U_{jk}|e^{i\theta_{jk}}$, we have $W_{jk} = |U_{jk}|\cos(\theta_{jk} + \phi_k)$, so $|W_{jk}| \leq |U_{jk}|$ and $\sum_k W_{jk}^2 \leq \sum_k |U_{jk}|^2 = 1$ for every row, whence $\|Wx\|_2 \leq \|x\|_2$: mixing never amplifies energy, while individual weights become negative whenever a phase shifts by π (destructive interference).*

Theorem 2 (Exact subtractive disentanglement). *There exist parameters of Definition 1 such that row 0 of W equals $\frac{1}{\sqrt{2}}(1, -1, 0, \dots, 0)$ while S remains doubly stochastic; with readout gain $\sqrt{2}$ the layer computes $X_0 - X_1$ exactly.*

Proof. Take A with a single block $\begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$, $a = \tan \frac{\pi}{4}$, $B = 0$; the Cayley transform of a rotation generator is the rotation with $\tan \frac{\theta}{2} = a$, i.e. $\theta = \frac{\pi}{2}$, and row 0 of $U = R(\theta) \oplus I_{n-2}$ is

Table 1: **Operator benchmark** ($n = 4$): worst doubly-stochastic violation of S over 100 random draws, and 1000 forward+backward passes at $(B, T, d) = (32, 256, 384)$.

Operator	row err	col err	params	ms/iter	peak MB
UniMHC (Ours)	4.8e-07	7.2e-07	36	10.56	352
GivensMHC (Ours-Fast)	3.6e-07	3.6e-07	10	10.85	352
Sinkhorn mHC ($T=20$)	1.2e-07	5.0e-03	16	12.93	352
Orthostochastic (NS)	2.4e-07	2.4e-07	16	11.23	352

$\frac{1}{\sqrt{2}}(1, 1, 0, \dots, 0)$ (sign of a chosen accordingly). Set $\phi = (0, \pi, 0, \dots, 0)$. Then $W_{00} = \text{Re}(U_{00}) = \frac{1}{\sqrt{2}}$ and $W_{01} = \text{Re}(U_{01}e^{i\pi}) = -\frac{1}{\sqrt{2}}$; the other entries of row 0 vanish. Doubly stochasticity of $S = |U|^{\odot 2}$ holds by Theorem 1 for every parameter value, including this one. $\square$

Proposition 2 (Impossibility floor for non-negative mixers). *Let $X_0, \dots, X_{n-1}$ be i.i.d. zero-mean unit-variance and $Y = X_0 - X_1$. For any readout c whose entries share a sign—the cone induced by every entrywise non-negative mixing row, including Sinkhorn/Birkhoff projections and orthostochastic readouts $Q \odot Q$ —,*

$$\min_{c \text{ same sign}} \mathbb{E} \left\| \sum_k c_k X_k - Y \right\|^2 = 1 = \frac{1}{2} \text{Var}(Y). \quad (3)$$

Proof. With $t = (1, -1, 0, \dots, 0)$, $\mathbb{E} \left\| \sum_k c_k X_k - Y \right\|^2 = \|c - t\|_2^2$ by independence and unit variance. The minimizer $c^* = t$ has mixed signs. On the all-nonnegative branch, coordinate-wise minimization under $c \geq 0$ clips the negative entry: $c = e_0$ with $\|e_0 - t\|^2 = 1$. On the all-nonpositive branch, $(c_0 - 1)^2 \geq 1$ already gives $\|c - t\|^2 \geq 1$. Hence the floor is 1. $\square$

The non-negativity of S buys exact energy conservation but forbids subtraction (floor $\frac{1}{2} \text{Var}(Y)$); the heterodyne readout keeps the energy budget unistochastic while releasing the sign of the mixing—probabilities are additive and non-negative, whereas amplitudes interfere. An inverse-free special case replaces U by $\frac{n(n-1)}{2}$ Givens rotations Q ($S = Q \odot Q$), a $O(n^2)$ -parameter unistochastic subfamily (Sec. 3.1).

3 Empirical Results

All experiments: NVIDIA RTX 4060 Laptop GPU (8 GB), PyTorch 2.x, WSL Ubuntu 24.04, stream width $n = 4$.

3.1 Operator precision and speed

Table 1: Uni-mHC and the Givens variant satisfy the constraint to $\sim 10^{-7}$ (pure float32 rounding), whereas Sinkhorn–Knopp with $T = 20$ leaves a column-sum residual of $5.0e - 03$ —the price of an iterative projection. The Cayley solve is also the cheapest: one well-conditioned 4×4 complex solve instead of 20 normalization sweeps (Thm. 1iv).

3.2 Subtractive disentanglement

We generate fresh $X \in \mathbb{R}^{512 \times 4 \times 64}$, $\mathcal{N}(0, 1)$, per step and ask the mixing layer alone (shared learnable gain γ) to produce $\hat{y} = \gamma \sum_k W_{0k} X_k$ against $Y = X_0 - X_1$; 800 Adam steps. Figure 2: Uni-mHC drives the loss to $1.14e - 11$ and GivensMHC to $9.72e - 15$ with coefficients $(+1.00, -1.00, 0, 0)$ —the exact destructive interference pattern of Theorem 2. Sinkhorn mHC and the orthostochastic baseline stall at $9.86e - 01$ and $9.85e - 01$ (floor: 1.0), their best solution being the trivial copy X_0 . All gradients were finite throughout, confirming stable complex autograd.

3.3 Language modeling with four residual streams

We modify nanoGPT (Karpathy, 2023): each of the 6 blocks (`n_embd=384`, 6 heads, context 256) operates over $n = 4$ residual streams; every attention/MLP stage reads a learned combination

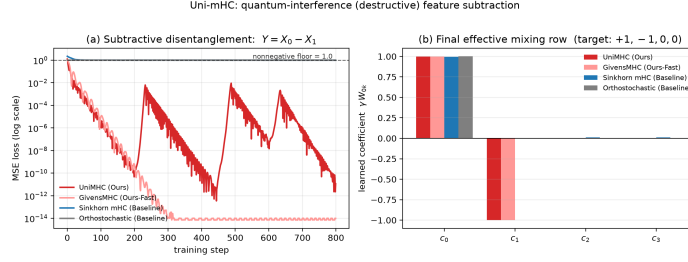


Figure 2: **Subtractive disentanglement** ($Y = X_0 - X_1$). **(a)** Uni-mHC and GivensMHC reach machine precision; non-negative baselines plateau at the impossibility floor (Prop. 2). **(b)** Learned coefficients γW_{0k} : phase interference yields the exact $(+1, -1, 0, 0)$ pattern; non-negative mixers degenerate to a pure copy $(+1, 0, 0, 0)$.

Table 2: **nanoGPT on shakespeare_char** (6 layers, 4 streams, 1000 steps). Val loss: mean of 50 eval batches; peak memory < 2.3 GB.

Mixer	final train loss	final val loss	ms/iter
vanilla GPT (no HC)	0.112	3.459	2644
Uni-mHC (Ours)	1.248	1.513	384
GivensMHC (Ours-Fast)	1.251	1.523	383
Sinkhorn mHC	1.259	1.507	546
Orthostochastic	1.249	1.505	483

$\sum_k a_k x_k$, writes back with learned per-stream weights, and a manifold mixer redistributes the streams, $x^{(t+1)} = M^{(t)} x^{(t)} + b^{(t)} \otimes f_t(\sum_k a_k^{(t)} x_k^{(t)})$. Mixers start at identity, so training begins as a standard pre-norm GPT (batch 32, 1000 steps, AdamW, cosine schedule, shakespeare_char; all runs resumable). Table 2 reports final losses.

At this scale the vanilla GPT overfits within the short horizon (train 0.11, val 3.46), while every manifold-constrained variant—whose identity-initialized stream topology warms up effective capacity gradually—holds val ≈ 1.51 . Stability is the second point: 12 mixers $\times$ 1000 steps with signed weights never destabilized training; post-training energy matrices violate double stochasticity by at most 7×10^{-7} , and all 12 mixers discovered signed interference weights—the unistochastic manifold is safely exploitable.

4 Conclusion

Uni-mHC replaces the projected doubly stochastic constraint of mHC with an exact unistochastic parameterization via a complex Cayley transform, keeping exact energy conservation and a non-expansive mixer while the heterodyne readout restores subtractive feature arithmetic through phase- π destructive interference—impossible for entrywise non-negative mixers, as we prove, and verified empirically (10^{-11} vs. 0.99). Future work: token-conditional phases, scaling studies, and links to quantum circuit compilation.

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